AuraReflex – A3 Academic Poster: Layout & Design Specification

Publication-ready text, grid, and design system for Canva / PowerPoint. Everything in the **COPY-READY** blocks can be pasted straight into your poster. Everything marked **[DESIGN CUE]** is an instruction for you, not poster text.

1. Canvas & Design System

Format: A3 portrait, 297 × 420 mm (11.69 × 16.54 in). Margins 12 mm. Bleed 3 mm if sent to print. Export PDF at 300 DPI, CMYK for print.

Grid: 3 columns, 8 mm gutters, with a full-width header band (about 15% of height) and a full-width footer band (about 7%).

Theme: "Deep-space electronics." Dark navy background with neon-cyan and electric-amber accents, circuit-trace line art, glassmorphism-style content cards.

Role	Colour	Hex
Background	Midnight navy	#0A1128
Card surface	Slate blue (85% opacity)	#121C3A
Primary accent (Auditory / Buzzer)	Electric cyan	#00E5FF
Secondary accent (Visual / LED)	Signal red-orange	#FF4D4D
Highlight (record value)	Amber gold	#FFC107
Body text	Soft white	#E8EEF9
Muted text / captions	Cool grey	#9AA8C7

Colour rule for every graph and number: Cyan = Buzzer (auditory). Red-orange = LED (visual). Keep this consistent across the whole poster; it lets viewers decode results in under a second.

Typography (free Google Fonts):

Element	Font	Size at A3
Title "AuraReflex"	Orbitron Bold or Exo 2 ExtraBold	85–100 pt
Subtitle	Rajdhani SemiBold	32–36 pt
Section headings	Orbitron / Exo 2 SemiBold, cyan, uppercase	28–30 pt
Body text	Inter or Roboto	17–19 pt
Big-number callouts	Orbitron Bold	54–64 pt
Captions, references	Inter Regular	12–13 pt

Style details: 2 px cyan glow line under each section heading; rounded cards (8–10 px radius) with a subtle 1 px cyan border; faint PCB-trace pattern at 6–8% opacity in the header and footer only. Keep body text high contrast and never place it over the pattern.

2. Page Map (Visual Placement Overview)

Zone	Content
HEADER (full width)	Logo left, title and subtitle centre, QR code right; authors and department strip below
COLUMN 1 (left)	1. Abstract, 2. Objectives, 3. System Architecture (block diagram), 4. Hardware & Firmware
COLUMN 2 (centre, hero)	5. Experimental Results: the 4-panel dashboard, KPI callouts, and the key-finding banner
COLUMN 3 (right)	6. Software / Analytics Layer, 7. Applications, 8. Conclusion & Future Work, 9. References
FOOTER (full width)	Group members, institution, GitHub QR, acknowledgements

[DESIGN CUE] Make Column 2 about 10% wider than the other two (roughly 36% / 28% / 36% split is not needed; use 31% / 38% / 31%). The results are your strongest asset, so give them the visual centre and the largest figures.

3. HEADER – COPY-READY

[DESIGN CUE] Left: Amrita Vishwa Vidyapeetham logo (white or transparent version). Right: QR code to the GitHub repo with the label "Scan for Code & Data." Place a thin cyan waveform or pulse line running behind the title to suggest a reflex signal.

AURAREFLEX

Real-Time Visual (LED) & Auditory (Buzzer) Dual-Participant Reaction Time Measurement System

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4. COLUMN 1 – COPY-READY

4.1 ABSTRACT

AuraReflex is a real-time, dual-participant biomedical reaction-time measurement system built as an ECE embedded-systems project. An **ESP32** microcontroller presents controlled **visual (LED)** or **auditory (buzzer)** stimuli, captures human responses with **microsecond-resolution timing** using polling and digital noise filtering, and streams serial telemetry over **UART** to a **Python** desktop application for automated CSV logging and 4-panel analytical plotting.

Trials with two participants over three rounds per stimulus showed that **auditory responses were consistently faster than visual responses**. Both participants showed clear **learning curves**, and the fastest single response recorded was **270.17 ms** (Participant 2, buzzer test).

[DESIGN CUE] Put the abstract in a glass card. Bold the three highlighted terms in cyan. Keep the card to about 130 words so it reads in 30 seconds.

4.2 OBJECTIVES

- Design a low-cost, portable, embedded platform for precise reaction-time measurement.
- Compare human response latency to **visual** vs. **auditory** stimuli under identical conditions.

- Support **two participants** with independent, simultaneous input channels.
- Automate data acquisition, statistics, logging, and visualisation end to end.

4.3 SYSTEM ARCHITECTURE

[DESIGN CUE: FIGURE 1 – BLOCK DIAGRAM] Draw a left-to-right flow with four coloured blocks joined by glowing arrows, as below. Use icons (LED, speaker, button, chip, laptop) inside each block.

```
[ STIMULUS LAYER ]  -->  [ FIRMWARE CORE ]  -->  [ UART TELEMETRY ]  -->  [
ANALYTICS LAYER ]
  Red LED                  ESP32 (C++/PlatformIO)     Serial stream
Python GUI
  Buzzer                   Timing + debounce          over USB
CSV + 4-panel plots
        ^                          ^
        |                          |
        +------ [ INPUT LAYER ] ---+
         Push button P1 (GPIO 32), P2 (GPIO 33)
```

Figure 1. AuraReflex system architecture: stimulus, input, firmware, and analytics layers.

4.4 HARDWARE & FIRMWARE

Layer	Implementation
Stimulus	ESP32 driving an external red LED (visual) and a 2-pin electromagnetic buzzer (auditory)
Input	Two tactile push buttons with internal pull-ups: P1 on GPIO 32, P2 on GPIO 33
Firmware	C++ on PlatformIO with a non-blocking 5 s countdown, a 7.0 s continuous active window, and digital debounce filtering
Link	UART serial telemetry to the host PC

[DESIGN CUE: FIGURE 2] Insert a photo of the assembled prototype (breadboard or PCB, ESP32, LED, buzzer, two buttons) with small callout labels. A clean top-down photo on a dark surface will match the theme. If a circuit schematic exists, place it as a thumbnail beside the photo.

5. COLUMN 2 (HERO) – COPY-READY

5.1 EXPERIMENTAL RESULTS

Protocol: 2 participants · 3 rounds per stimulus · LED and buzzer tested separately.

[DESIGN CUE: KPI STRIP] Directly under the heading, place four large number cards in a row (or a 2×2 grid if space is tight):

Card	Big number	Label	Colour
1	**270.17 ms**	Fastest response (P2, Buzzer)	Amber gold
2	**782.71 ms**	Buzzer average, P1	Cyan
3	**3698.36 ms**	LED average, P1	Red-orange
4	**3.8× – 4.7×**	Auditory faster than visual	White with cyan glow

[DESIGN CUE: FIGURE 3 – THE 4-PANEL DASHBOARD] This is the centrepiece. Two options, in order of preference:

1. **Best:** Export each panel from your Matplotlib dashboard as a separate high-resolution PNG (300 DPI, transparent or dark background) and arrange them in a 2×2 grid, so you control sizing and colours. Restyle Matplotlib with a dark theme, cyan for buzzer, red-orange for LED.
2. **Fast:** Place the full 4-panel screenshot as one wide image. Make sure the text inside it stays legible at A3 (axis labels at least 10 pt when printed).

Place the panels as follows, with the caption beneath each:

	Left	Right
Top row	**Panel A** – LED 3-Trial Reaction Curve	**Panel B** – Buzzer 3-Trial Reaction Curve
Bottom row	**Panel C** – Combined Average Response Time	**Panel D** – Combined Best Response Time

Panel A – Visual (LED) learning curve. Reaction time falls across trials: P1 from about 5000 ms to about 2300 ms, P2 from about 7000 ms to about 2700 ms.

Panel B – Auditory (Buzzer) learning curve. Responses are much faster and still improve: P1 from about 1400 ms to about 450 ms, P2 from about 2300 ms to about 580 ms.

Panel C – Average response time. Auditory processing is **3.8× to 4.7× faster** than visual.

Stimulus	Participant 1	Participant 2
LED (visual)	3698.36 ms	4765.75 ms
Buzzer (auditory)	782.71 ms	1248.18 ms

Panel D – Best response time. The project record is **270.17 ms**, set by Participant 2 in the buzzer test.

5.2 KEY FINDINGS BANNER

[DESIGN CUE] Full-width amber-bordered banner at the bottom of the column, large type (22–24 pt):

- **Auditory beats visual:** buzzer responses were faster than LED responses for both participants, in every comparison.
- **Practice matters:** both participants improved from trial to trial for both stimuli.
- **Record reflex: 270.17 ms** (Participant 2, buzzer).

6. COLUMN 3 – COPY-READY

6.1 SOFTWARE & ANALYTICS LAYER

A Python desktop application receives live UART telemetry and automates the whole analysis workflow.

- **PySerial:** real-time serial data acquisition
- **NumPy:** statistical aggregation (Average, Best, Standard Deviation)
- **Matplotlib:** automated 4-panel dashboard generation
- **CSV export:** one-click session logging for reproducible analysis

[DESIGN CUE: FIGURE 4] Insert a screenshot of the Python GUI or serial log, in a rounded frame with a faint cyan glow. It shows the system is real and working.

6.2 APPLICATIONS

[DESIGN CUE] Use a 5-row icon list: one small line icon (car, runner, brain, gamepad, factory) beside each item.

Icon	Application	Value
Car	**Driver Alertness & Fatigue Screening**	Assess cognitive latency before operating vehicles or heavy machinery
Runner	**Sports Science & Reflex Training**	Track athlete reflex improvement trial by trial
Brain	**Cognitive & Neurological Baseline Testing**	Low-cost, portable tool for reflex assessment
Gamepad	**Gaming & E-Sports Benchmarking**	Quantify auditory vs. visual reaction differences in competitive play
Factory	**Industrial Safety & HMI Design**	Test whether operators respond faster to visual or auditory alarms

6.3 CONCLUSION & FUTURE WORK

Conclusion. AuraReflex shows that a low-cost ESP32-based platform can deliver precise, repeatable, dual-participant reaction-time measurement with fully automated analytics. The results agree with established findings that auditory reaction times are shorter than visual ones [1–3].

Future work.

- Expand to larger participant groups for statistically robust conclusions
- Add wireless (BLE / Wi-Fi) telemetry and a mobile dashboard
- Add haptic stimulus and multi-stimulus (choice reaction) tests
- Add cloud logging and longitudinal tracking per user

6.4 REFERENCES

[DESIGN CUE] 12 pt, muted grey, tight line spacing.

[1] D. L. Woods *et al.*, "Factors influencing the latency of simple reaction time," *Frontiers in Human Neuroscience*, vol. 9, p. 131, 2015.

[2] J. Shelton and G. P. Kumar, "Comparison between auditory and visual simple reaction times," *Neuroscience & Medicine*, vol. 1, no. 1, pp. 30–32, 2010.

[3] R. J. Kosinski, "A literature review on reaction time," Clemson University, 2008.

[4] J. D. Hunter, "Matplotlib: A 2D graphics environment," *Computing in Science & Engineering*, vol. 9, no. 3, pp. 90–95, 2007.

[5] S. Uday Kumar *et al.*, "AuraReflex: Real-Time ESP32 Reaction Measurement System," GitHub Repository, 2026. [Online]. Available: https://github.com/udaysadaraboina/AuraReflex-ECE-Project

7. FOOTER – COPY-READY

[DESIGN CUE] Full-width darker band with a thin cyan top border. Left: QR code to the GitHub repo. Centre: team and institution. Right: department or university logo.

Team AuraReflex: S. Uday Kumar · R. Pradeep · G. Kushwant · B. Vamsi Krishna · B. Raja Reddy

Department of Electronics & Communication Engineering · Amrita Vishwa Vidyapeetham, Amritapuri Campus

Code & data: github.com/udaysadaraboina/AuraReflex-ECE-Project

Optional line: "Guided by: [Faculty Guide Name]" and "Acknowledgements: [Lab / Guide]." Fill these in if applicable.

8. Pre-Print Checklist & Expert Notes

Presentation tips

- **Reading order:** Title, KPI strip, 4-panel graphs, key findings. A passer-by should get the story in 10 seconds without reading a paragraph.
- **Whitespace:** keep at least 15% empty space; resist the urge to fill every gap.
- **Figures over words:** target roughly 40% visuals, 40% text, 20% whitespace.
- **Legibility test:** view the poster at 25% zoom on screen. The title, KPIs, and section headings must still read clearly.
- **Print:** use matte or satin paper. Dark backgrounds are more sensitive to banding, so print a proof of a small area first.

Accuracy notes (worth fixing before the exhibition)

1. **Sample size:** with 2 participants and 3 rounds per stimulus, this is a pilot study. The poster wording above says "showed" rather than "proved," and the conclusion points to larger groups as future work. This pre-empts the most likely judge question.
2. **"Microsecond precision":** the hardware timer resolves microseconds, but results are reported in milliseconds. The poster says "microsecond-resolution timing" to stay accurate.
3. **The 3.8×–4.7× ratio checks out:** 4765.75 ÷ 1248.18 ≈ 3.82 (P2) and 3698.36 ÷ 782.71 ≈ 4.73 (P1).
4. **Likely judge questions to prepare for:** why are first-trial values so high (about 5–7 s for LED)? What is the 7 s window logic for missed responses? How is debounce implemented, and what delay does it add? Have a one-sentence answer ready for each.
5. **Panel values** in section 5 (such as "about 5000 ms") are read from your graph. Before printing, replace them with exact per-trial values from your CSV for maximum credibility.

Export workflow (Canva / PowerPoint)

- **Canva:** Custom size 297 × 420 mm. Create 3 column guides, paste each section's COPY-READY text into separate rounded rectangles, and upload your graph PNGs. Download as PDF Print.
- **PowerPoint:** Design tab, Slide Size, Custom, 29.7 × 42.0 cm, Portrait. Use Insert, Table or rounded rectangles for cards. File, Export, PDF, Standard (print quality).